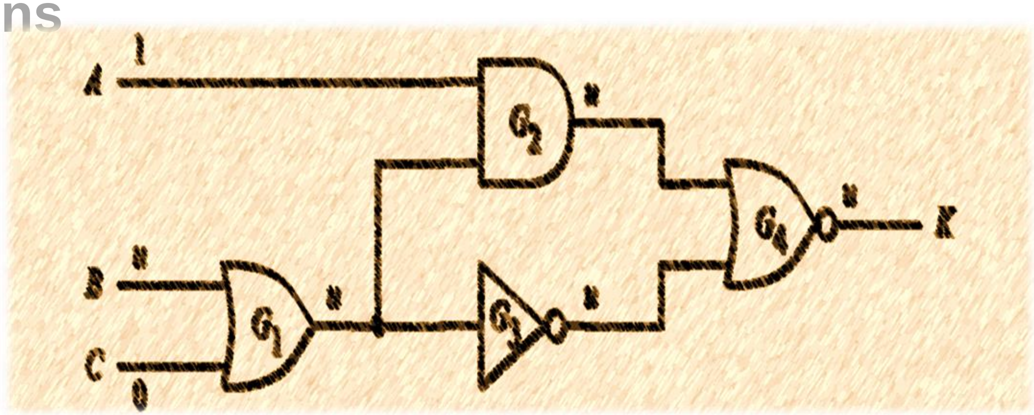
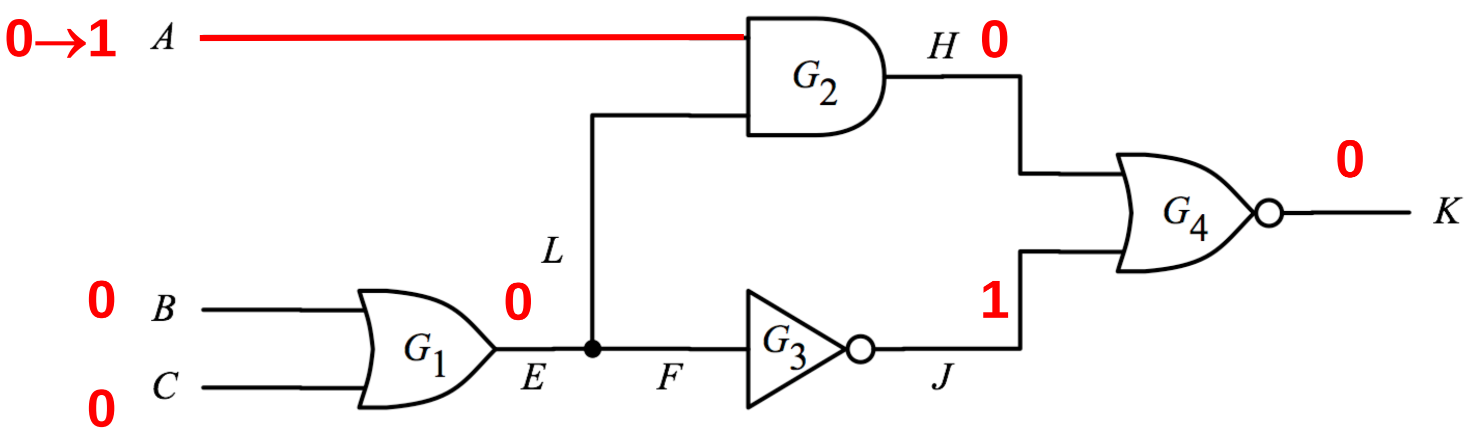
Logic Simulation

- Introduction
- Simulation Models
- Logic Simulation Techniques
  - Compiled-code simulation
  - Event-driven simulation (1965)
    - Zero delay
    - Nominal delay
    - False events
  - Parallel Simulation
- Issues of Logic Simulations
- Conclusions



Compiled-code Simulation Problems

- 1. Gate delay model not considered
- 2. *Oblivious:* forgets results in previous cycle




```
while{true} do
        read(A,B,C);
        E   OR(B,C);
        H   AND(A,E);
        J   NOT(E);
        K   NOR(H,J);
end
```

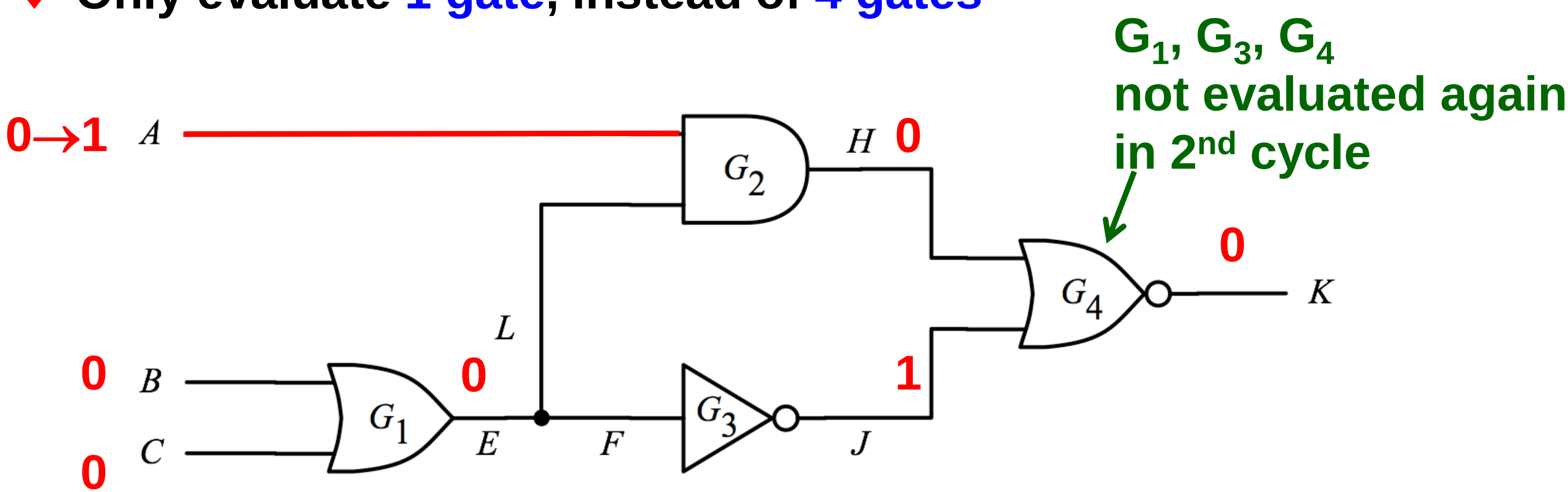
1^{st} 2^{nd}

Forget 1^{st} cycle results.

Rerun all codes for 2^{nd} cycle.

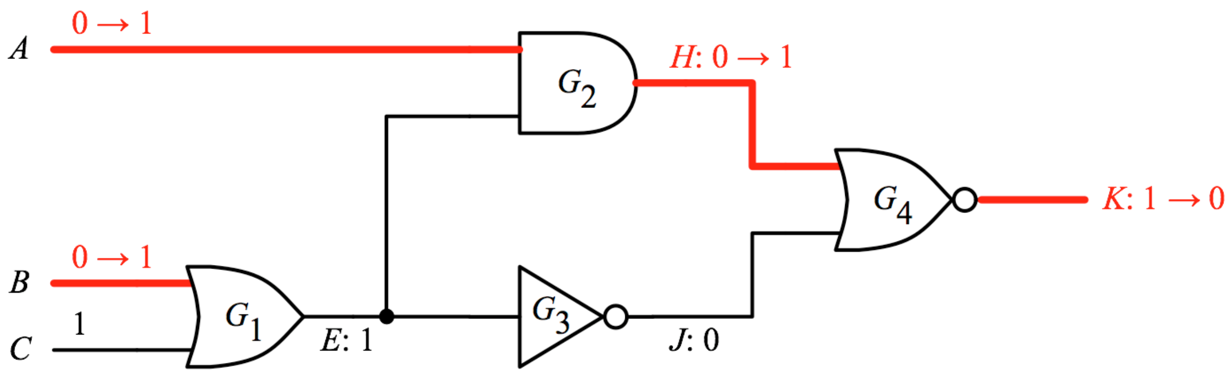
Event-Driven Simulation [Ulrich 1965]

- IDEA: evaluates a gate only if there is event(s) at its gate inputs
  - *Event* is a signal value change (at time *t*)
  - Gates with inputs changed are *activated*
- Example:
  - Event: *A* 0→1
  - Activated gate: G_2
  - Only evaluate 1 gate, instead of 4 gates



Event-driven Faster then CC

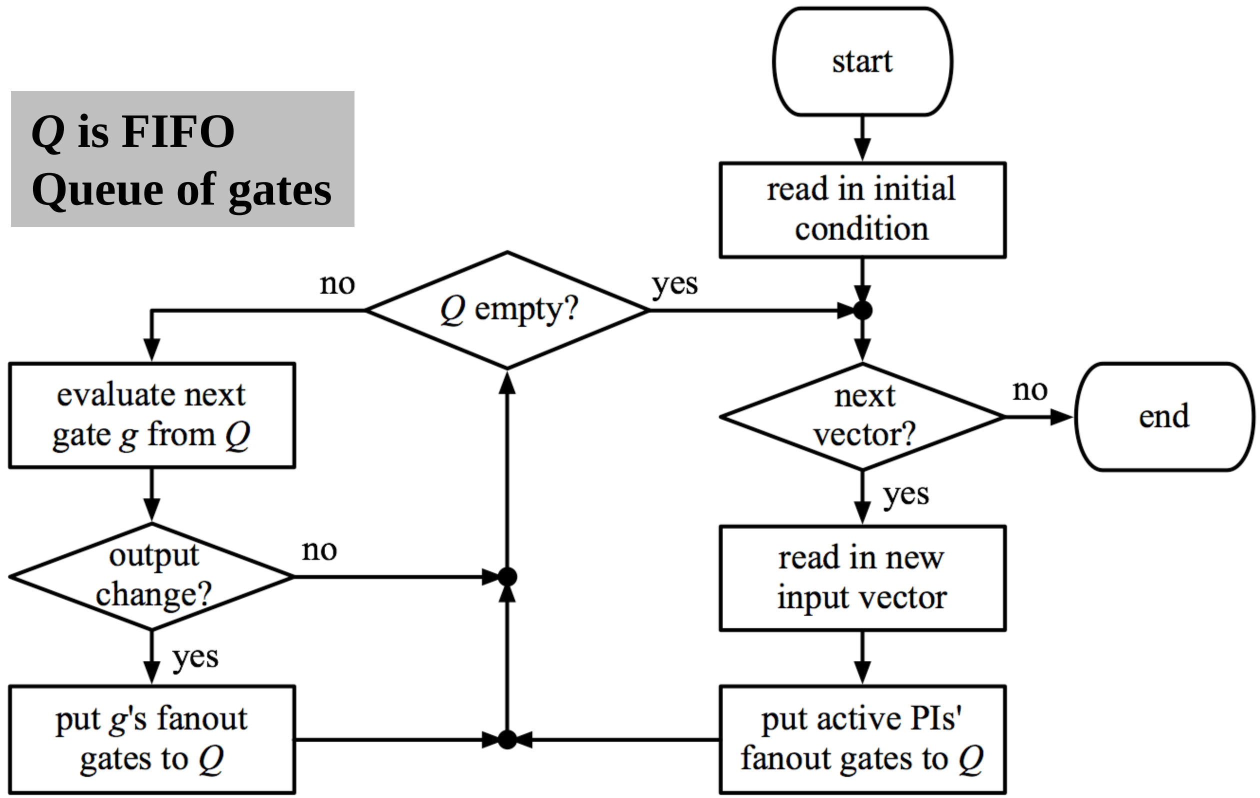
Zero-delay Event-driven Sim.



event (g, v_g^+) **means gate *g* changes to v_g^+**

Q **is FIFO Queue of gates**

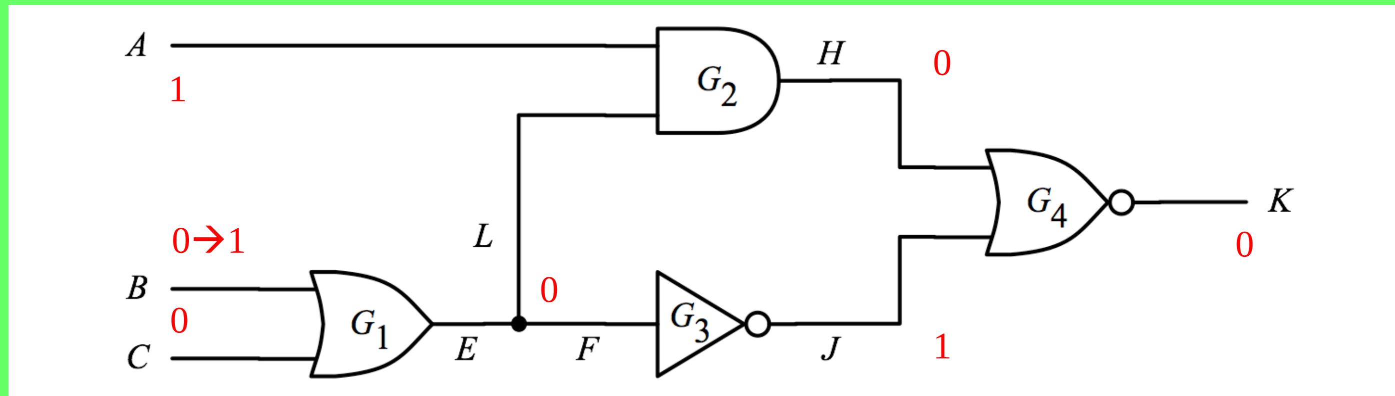
Executed events	Q
(*B*,1)(*A*,1)	G_1, G_2
(G_1, 1) -	G_2
(G_2,1)	G_4
(G_4,0)	-



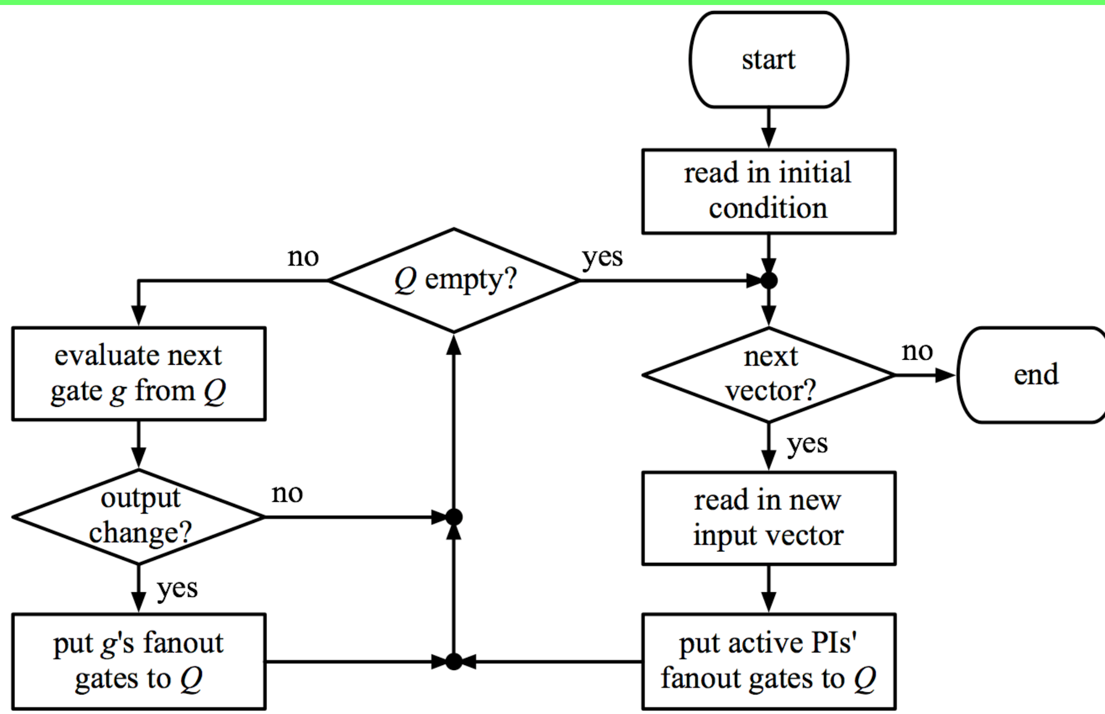
(WWW Fig 3.16)

Quiz

Please simulate this circuit



Executed Events	Q
(*B*,1)	G_1



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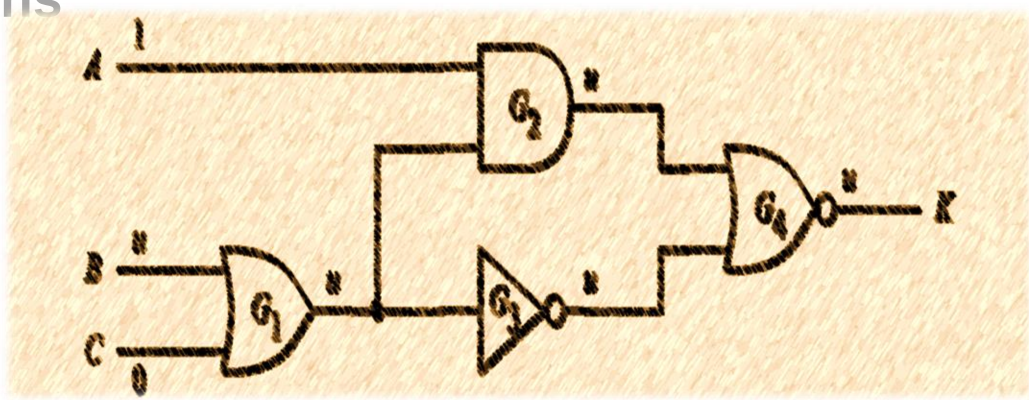
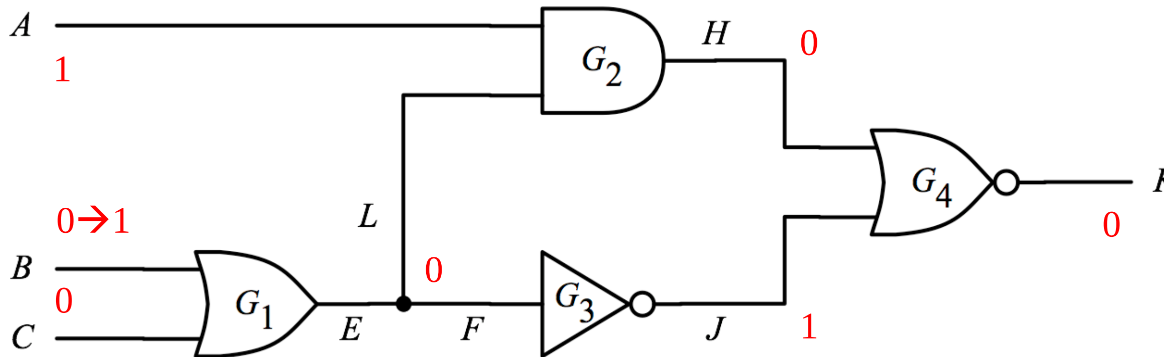


Nominal-delay Event-drive Sim.

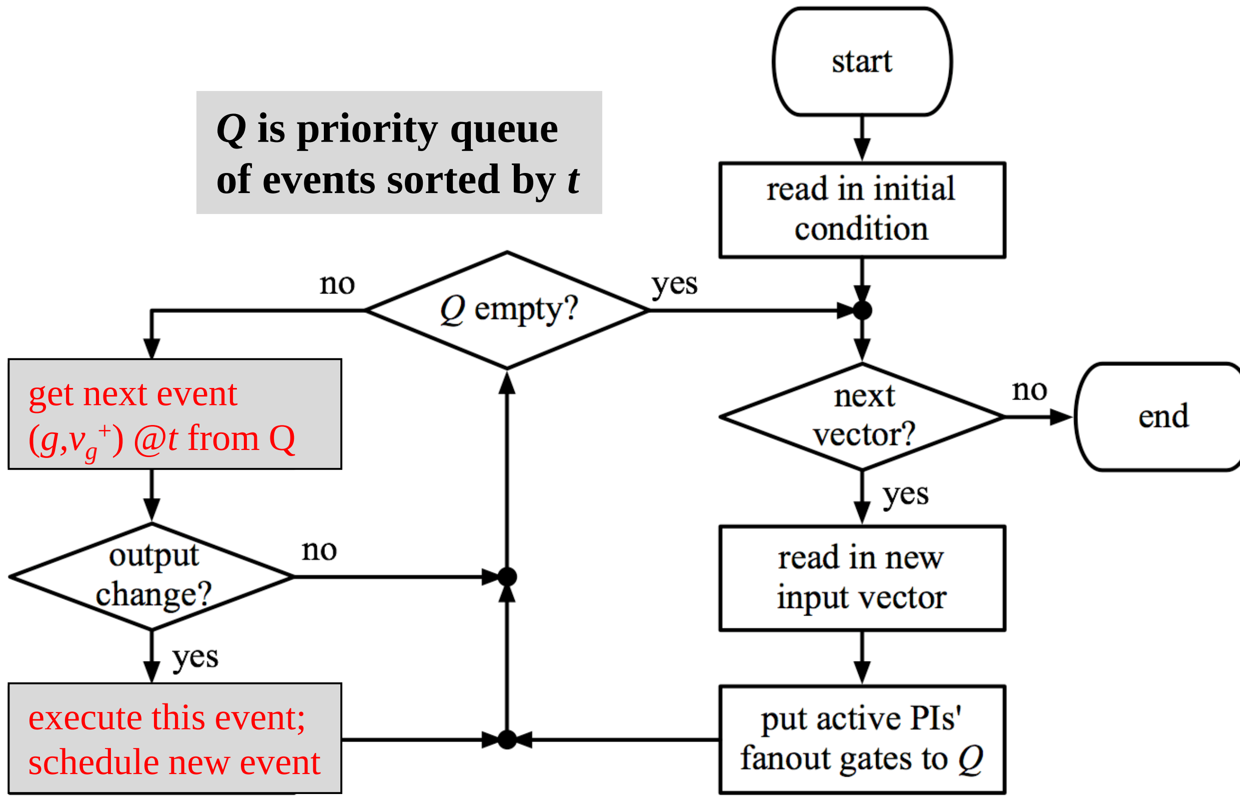
every gate delay = 1ns

event (g, v_g^+) @ *t*
gate *g* changes to v_g^+
at *time stamp t*



t	Executed events	sch'd events Q
0	$(B,1)$	$(G_1,1)$@1
1	$(G_1,1)$	$(G_3,0)$@2 $(G_2,1)$@2
2	$(G_3,0)$	$(G_2,1)$@2 $(G_4,1)$@3
2	$(G_2,1)$	$(G_4,1)$@3 $(G_4,0)$@3
3	$(G_4,1)$	$(G_4,0)$@3
3	$(G_4,0)$	

Q is priority queue of events sorted by *t*

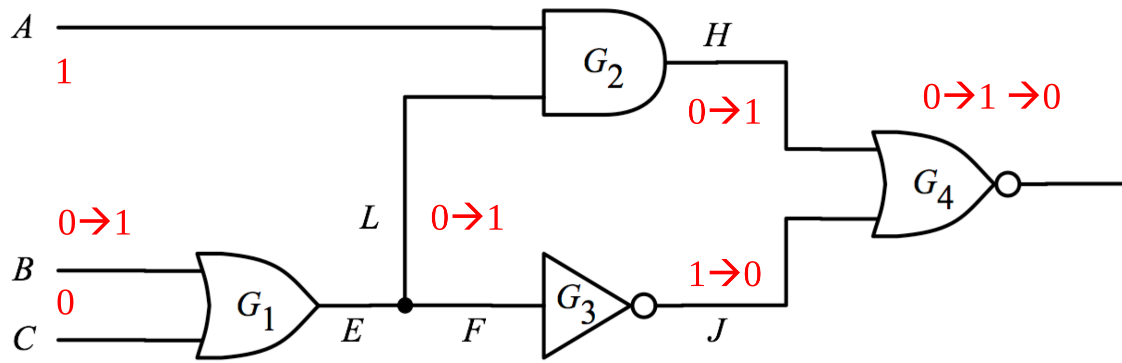


FFT

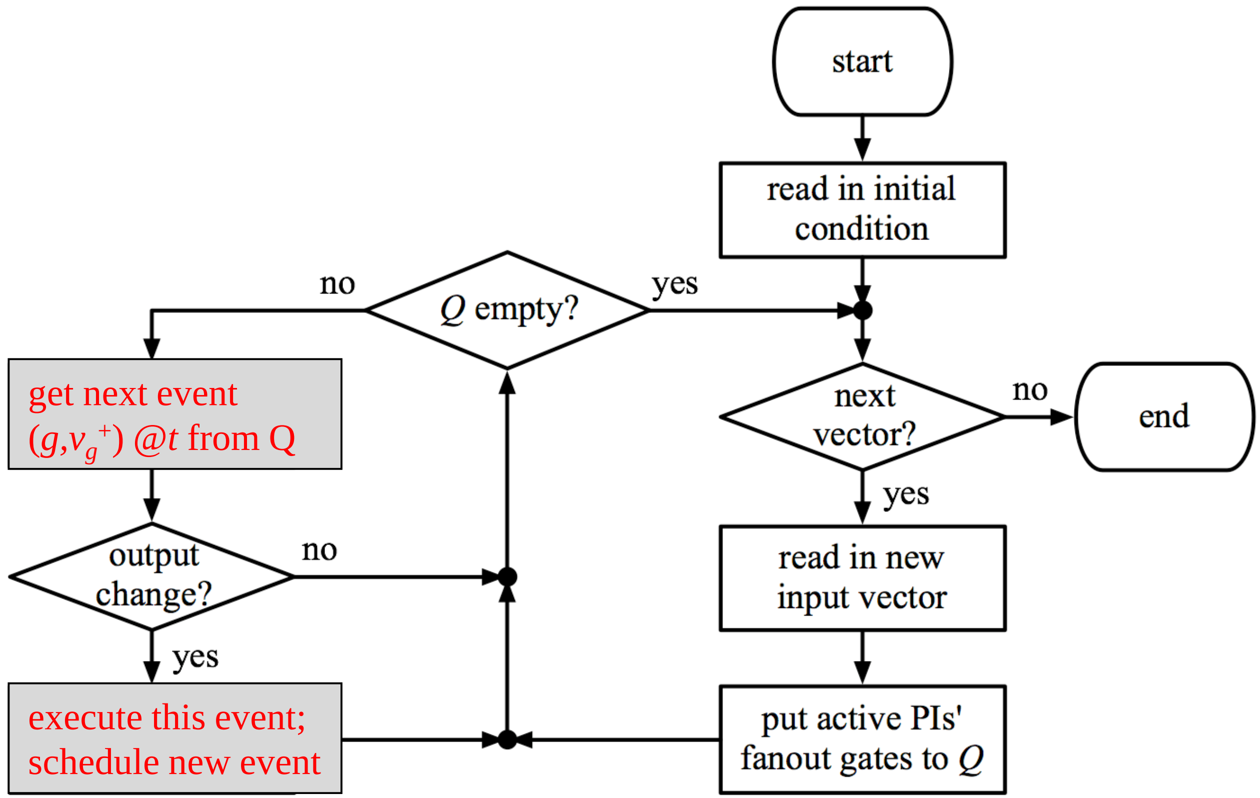
Simultaneous signal change

of G_4 @3 wastes CPU time.

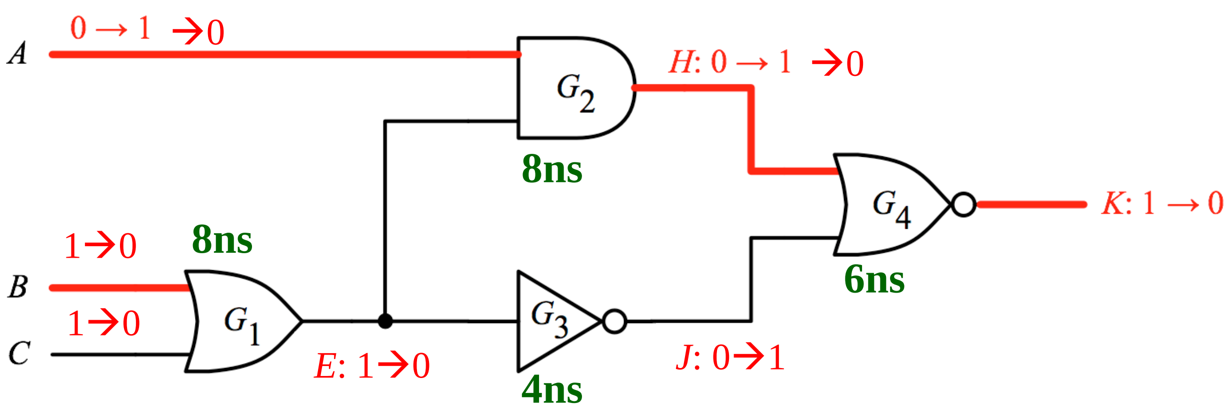
Q: Can we do better?



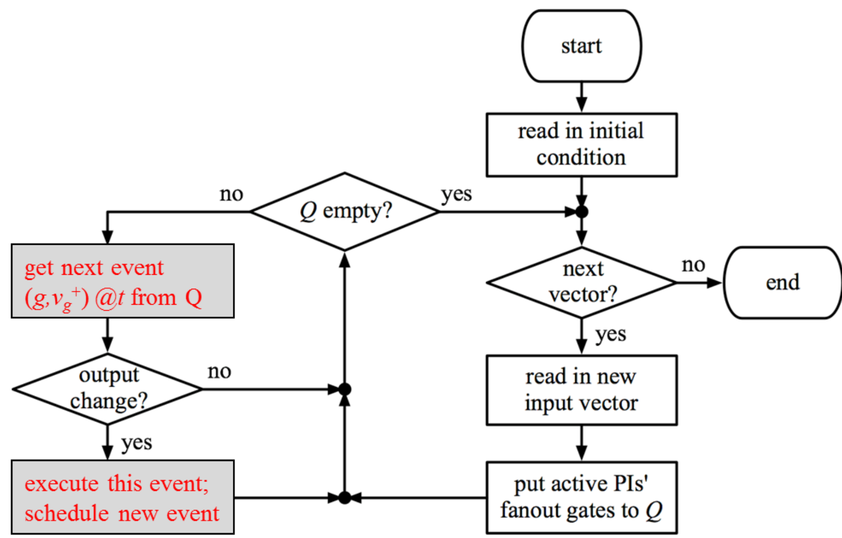
t	Executed events	sch'd events Q
0	$(B,1)$	$(G_1,1)$@1
1	$(G_1,1)$	$(G_3,0)$@2 $(G_2,1)$@2
2	$(G_3,0)$	$(G_2,1)$@2 $(G_4,1)$@3
2	$(G_2,1)$	$(G_4,1)$@3 $(G_4,0)$@3
3	$(G_4,1)$	$(G_4,0)$@3
3	$(G_4,0)$	



Quiz



t	executed	scheduled events in Q
0	(A,1)	(G_2,1)@8
2	(C,0)	(G_2,1)@8; (G_1,1)@10
4	(B,0)	(G_2,1)@8; (G_1,1)@10; (G_1,0)@12
8	(A,0),(G_2,1)	(G_1,1)@10; (G_1,0)@12 (G_4,0)@14; (G_2,0)@16
10	(G_1,1)	(G_1,0)@12;(G_4,0)@14;(G_2,0)@16
12	(G_1,0)	(G_4,0)@14; (G_2,0)@16 (G_3,1)@16; (G_2,0)@20
14	(	
16	(	
20	(	
22	(G_4,1)(G_4,0)(G_4,0)	-



*Two events are shown in same row for simplicity

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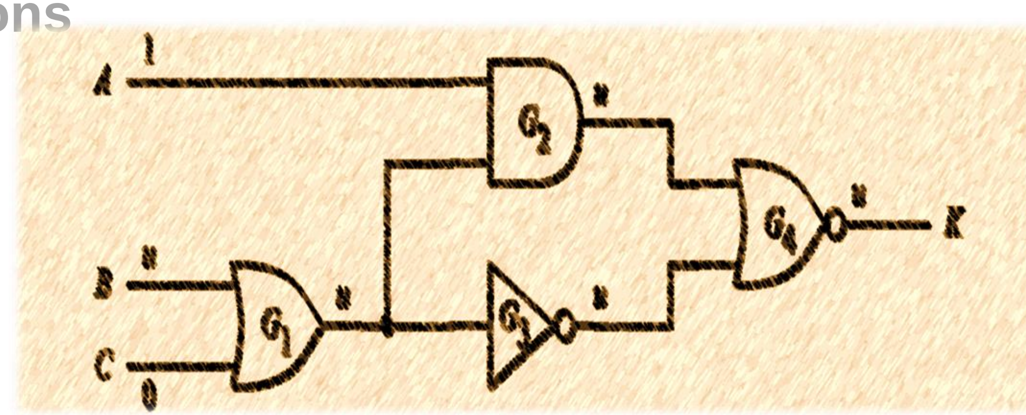
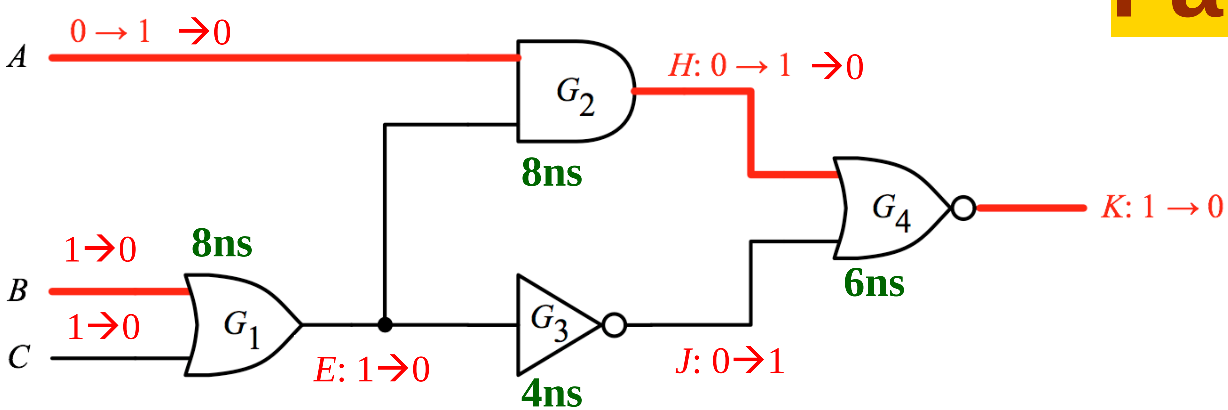
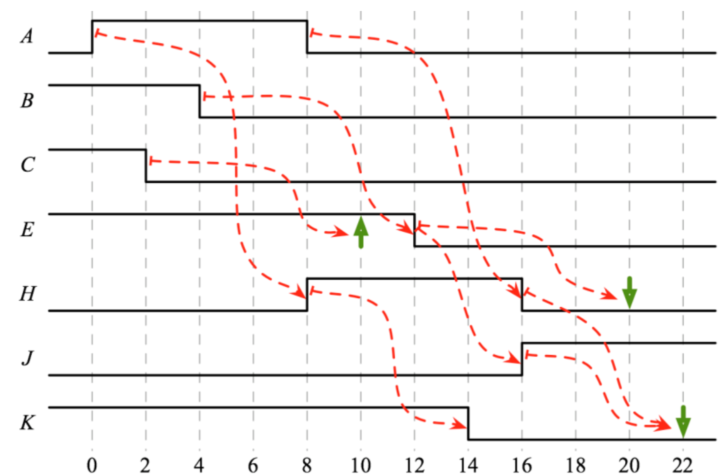


False Events

A false event refers to a scenario where an event or signal transition is processed during the simulation but ultimately has no effect on the circuit's state or behavior. These events occur due to temporary or redundant changes in signal values that resolve themselves before influencing downstream logic.



t	executed	scheduled events in Q
0	$(A,1)$	$(G_2,1)@8$
2	$(C,0)$	$(G_2,1)@8$; $(G_1,1)@10$
4	$(B,0)$	$(G_2,1)@8$; $(G_1,1)@10$; $(G_1,0)@12$
8	$(A,0),(G_2,1)$	$(G_1,1)@10$; $(G_1,0)@12$ $(G_4,0)@14$; $(G_2,0)@16$
10	$(G_1,1)$ false	$(G_1,0)@12$;$(G_4,0)@14$;$(G_2,0)@16$
12	$(G_1,0)$	$(G_4,0)@14$; $(G_2,0)@16$ $(G_3,1)@16$; $(G_2,0)@20$
14	$(G_4,0)$	$(G_2,0)@16$;$(G_3,1)@16$;$(G_2,0)@20$
16	$(G_2,0)(G_3,1)$	$(G_2,0)@20$; $(G_4,1)@22$; $(G_4,0)@22$
20	$(G_2,0)$ false	$(G_4,1)@22$; $(G_4,0)@22$; $(G_4,0)@26$
22	$(G_4,1)(G_4,0)(G_4,0)$	-



***False Events* cause no change in signal**

Event-driven Alg. w/o False Events

- **For each gate, record *last scheduled value* to avoid false events**
- **(BA Fig. 3.32)**

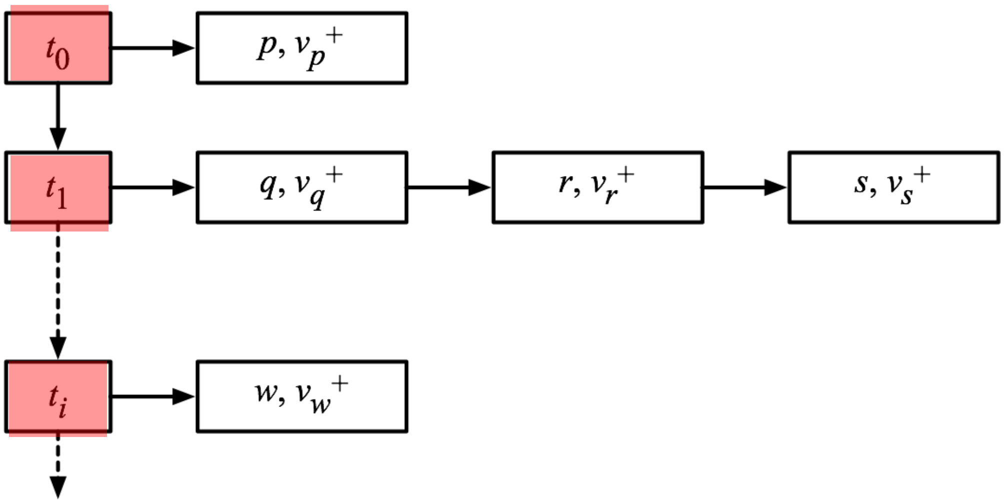
event-driven-sim (t) /* t is current time stamp*/

for every event (g, v_g^+) @ t
    $v_g = v_g^+$
    for every j on the fauout list of g
        update input values of j
        v_j^+ = evaluate (j)
        if $v_j^+ \neq \mathrm{lsv}(j)$ /* lsv = last scheduled value */
            schedule (j, v_j^+) @ $t+d_j$ /* d_j = *gate delay of j* */
            $\mathrm{lsv}(j) = v_j^+$

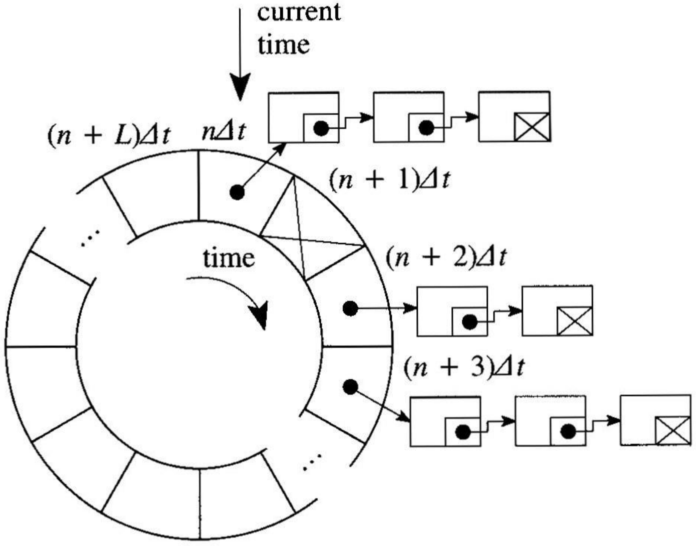
Trade off Memory for Speed

How to Implement Event List ?

1. Linked list
   - ☺ *t* is **flexible**
   - ☹ **Slower** to search

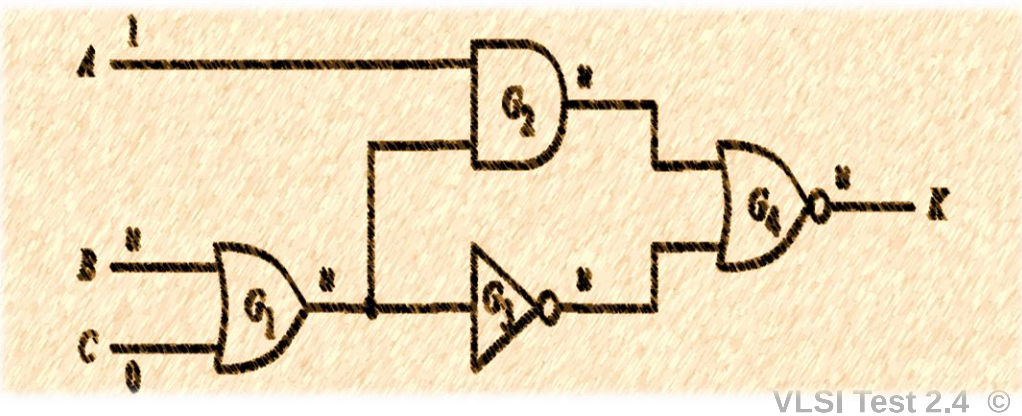


2. Cyclic Array *(aka. **Timing Wheel**)* [Ulrich 1969]
   - ☺ **Faster** to search
   - ☹ Δt is **fixed**
   - ☹ Limited L slots in a cycle



Summary

- Event-driven simulation
  - Consider gate delay
  - Saves CPU time. Only evaluate gates with events at input.
- Two algorithms
  - Zero delay
  - Nominal delay
- False events are redundant events that can be removed
- Event list implementation
  - Linked list: slow but flexible
  - Cyclic Array (Timing wheel): fast but fixed time slots



Comparison

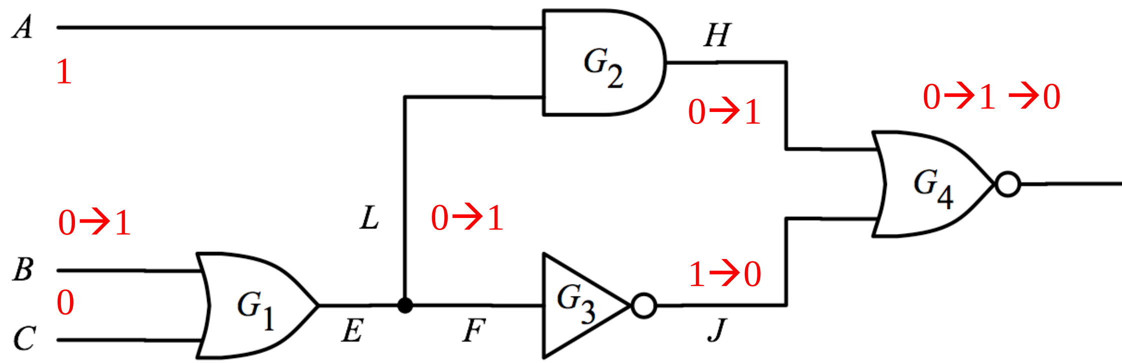
	Pros ☺	Cons ☹
Compiled-code	**Simple implementation**	**No gate delay** **Oblivious** **Suitable for high activity**
Event-driven	**Consider gate delay** **Only simulate events** **Suitable for low activity**	**Complex algorithm**

ED is Generally Better than CC

FFT

Simultaneous signal change **of G_4 @3 wastes CPU time.**

Q: Can we do better?



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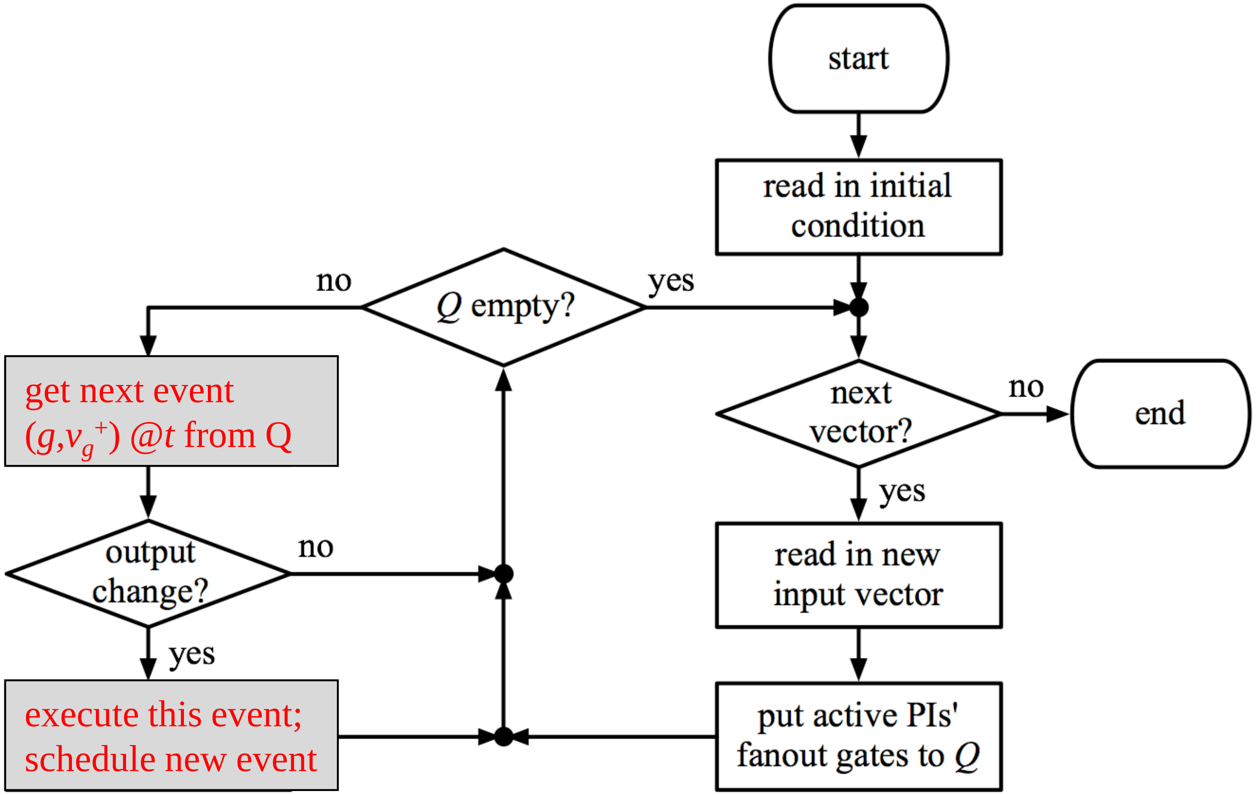
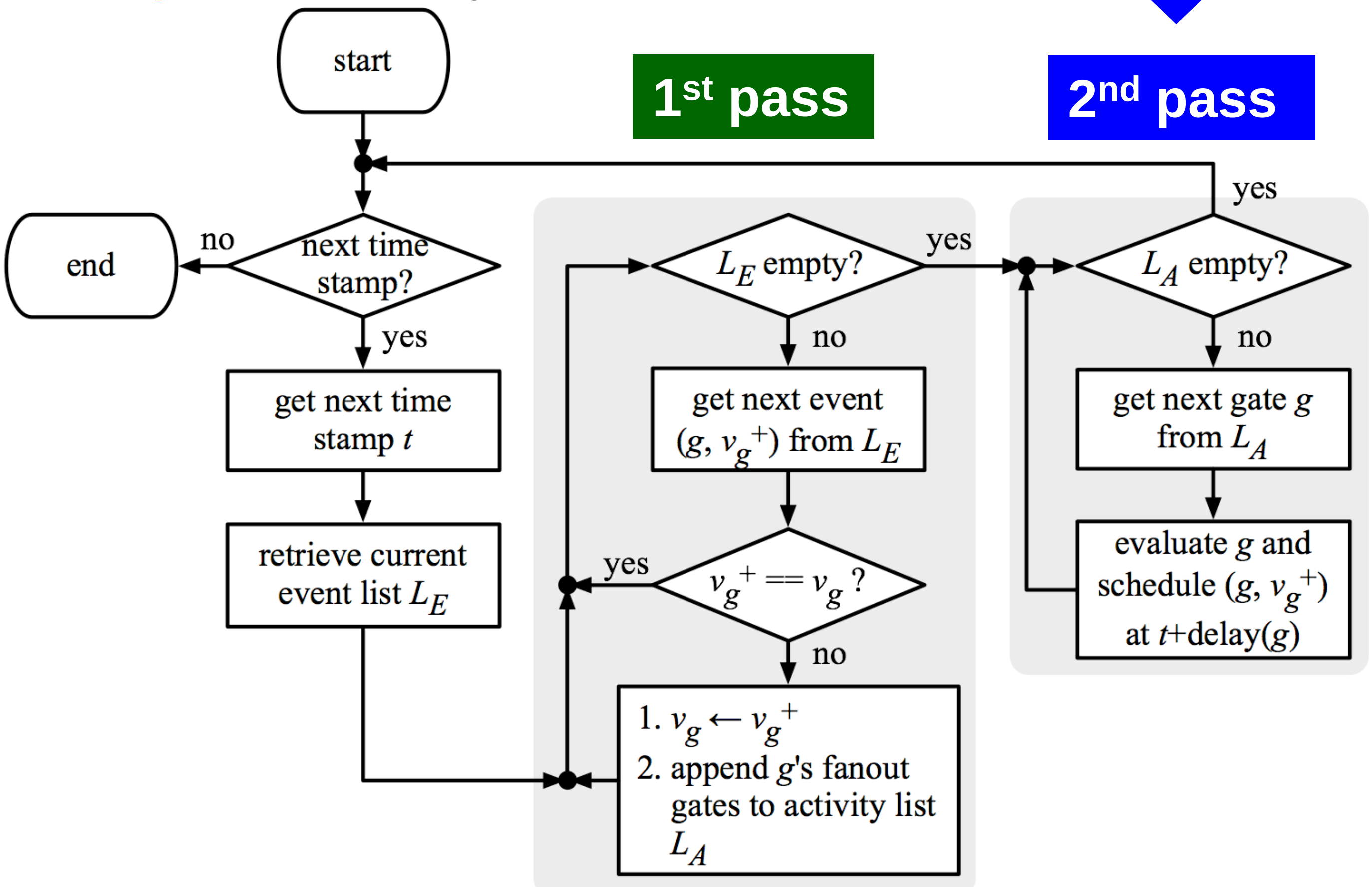
1	(G_1,1)	(G_3,0)@2 (G_2,1)@2
2	(G_3,0)	(G_2,1)@2 (G_4,1)@3
2	(G_2,1)	(G_4,1)@3 (G_4,0)@3
3	(G_4,1)	(G_4,0)@3
3	(G_4,0)	



IDEA#1 Two-pass Algorithm [Ulrich 1965]

- 1st pass executes events
  - L_E event list, priority queue of events
- 2nd pass evaluates gates
  - L_A activity list, list of gates

All input events to a gate evaluated together.
Avoid simultaneous signal change



IDEA #2 One-pass Algorithm [Ulrich 1969]

- **Add last scheduled time (lst)**
- **Cancel previously scheduled events when needed**

event-driven-sim-new (t)

for every event (g, v_g^+) @ t

$v_g = v_g^+$

for every j on the fauout list of g

update input values of j

v_j^+ = evaluate (j)

if $v_j^+ \neq \mathrm{lsv}(j)$

$t^+ = t + d_j$

if $t^+ == \mathrm{lst}(j)$ /* simultaneous signal change*/

cancel event (j, lsv(j)) @ t^+

schedule (j, v_j^+) @ t^+

lsv(j) = v_j^+

lst(j) = t^+ /* lst = last scheduled time */

FFT

- **Cyclic Array *(aka. Timing Wheel*)**
  - ☹ **Limited L slots in a cycle**
- **Q:what if *remote events* that is outside of $(n+L)\Delta t$?**